Erika Alexandra García Barrios

Software Engineer · AI & Machine Learning · Java Backend · Web Development

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PROFESSIONAL SUMMARY

Software Engineer focused on AI systems, Machine Learning, backend and web development. I build LLM, RAG and agent systems that run in production, deep learning models for medical imaging, backend services in Java and Python, and custom web applications with React, Next.js and TypeScript, from client e-commerce to institutional platforms. Experience in the public and private sectors with Java, Spring Boot, REST and SOAP APIs and relational databases, applying clean architecture, SOLID principles and automated testing. Approved thesis on breast cancer classification in DCE-MRI (ROC-AUC 0.941), now being prepared as a scientific paper. Expected graduation: December 2026.

EXPERIENCE

AI Systems & Automation Engineer — Independent / Freelance — 2023 — Present

Design and development of AI systems with LLMs, RAG and agents, plus custom web development for clients, integrated through APIs, web interfaces and containers.

- Vella Perfumería (client): custom e-commerce with Next.js 16, React 19, TypeScript and Tailwind, and a Spring Boot 4 backend with Spring Security, JPA, Flyway and PostgreSQL; real catalog of more than 100 products, cart, payments and admin panel. Storefront live on Vercel.
- Montajes y Servicios García (client): bilingual (Spanish/English) landing page with light and dark themes, responsive design and direct WhatsApp contact, deployed on Vercel.
- SaberPro-RAG: RAG study assistant covering the five modules of Colombia's Saber Pro exam (Qdrant, sentence-transformers, Ollama / Mistral), documented with ADRs.

Python · FastAPI · React · Next.js · TypeScript · Spring Boot · PostgreSQL · Docker · LLMs · RAG · AI Agents · Redis · Vercel

Data Science & Machine Learning Researcher — CognIA Research Group — Institución Universitaria Pascual Bravo — Feb 2026 — Present

Predictive models to classify benign and malignant breast tumors with the Breast Cancer Wisconsin Diagnostic dataset.

- Compared logistic regression, decision trees, random forest, SVM and KNN with K-Fold cross-validation, stratified splits and hyperparameter tuning.
- Exploratory analysis, cleaning, transformation and feature selection with descriptive statistics and correlation analysis.
- Evaluation with confusion matrices, accuracy, precision, recall, specificity, F1, ROC and AUC.

Python · Scikit-learn · Pandas · NumPy · Matplotlib · Seaborn · Transformers · CNN · SVM

AI & Computer Vision Researcher — Undergraduate thesis — Institución Universitaria Pascual Bravo — Sep 2025 — Jun 2026

Deep learning to classify benign and malignant breast lesions in DCE-MRI. Thesis approved in September 2026; now being turned into a scientific paper with the guidance of Dr. Rubén Darío Fonnegra Tarazona.

- Compared ResNet50, EfficientNet-B3 and MobileViT-S with transfer learning and progressive fine-tuning, K-fold cross-validation during training and a final patient-level hold-out evaluation (BreastDM).
- Final ResNet50 model on the test set (35 patients): ROC-AUC 0.941, sensitivity 0.864 (19 of 22 malignant), specificity 0.846 and F1 0.884; calibrated AUC (isotonic) 0.897, 95% CI [0.766, 0.992].
- Interpretability with Grad-CAM; experiment tracking with MLflow and TensorBoard.

Python · PyTorch · OpenCV · ResNet50 · EfficientNet-B3 · MobileViT-S · Grad-CAM · MLflow · TensorBoard

Webmaster · Institutional Web Development (contractor) — Institución Universitaria Pascual Bravo — Jul 2025 — Oct 2025

Management and development of the university's web platforms and support for digital transformation initiatives.

- Maintained, updated and optimized institutional web platforms, improving responsive design, usability and user experience.
- Automations with Google Apps Script.

HTML · CSS · JavaScript · Google Apps Script · WordPress · Moodle

Junior Full Stack Developer — Java Backend — Servisoft S.A. Jun 2023 — Sep 2024

Custom Java backend solutions for public and private sector clients: DANE, JEP, Medellín City Hall, Medellín District Comptroller's Office, Bucaramanga Metropolitan Water Utility, UPB, Pascual Bravo and Viva.

- Backend development in Java 8/12 (Java SE, JSP, Servlets, JDBC) on Apache Tomcat 6–9, with REST and SOAP APIs for critical applications.
- MSSQL, Oracle and MongoDB: CRUD operations, stored procedures and query optimization.
- Diagnosed, fixed and modernized legacy systems in Java and JavaScript, providing technical and functional support.

Java 8/12 · JSP · Servlets · JDBC · Apache Tomcat · REST · SOAP · MSSQL · Oracle · MongoDB · Git

Software Development Intern — MV-TEL SAS Jun 2021 — Nov 2021

Full development cycle of a mobile app and the corporate website within a Scrum team.

- Requirements gathering and analysis, functional design, architecture definition and feature validation.
- Responsive corporate website with HTML, CSS and JavaScript.

Java · Selenium WebDriver · HTML · CSS · JavaScript · Scrum

RESEARCH AND PUBLICATIONS

- Scientific paper (in preparation) derived from the DCE-MRI breast lesion classification thesis.
- Speaker at the 21st International Energy Symposium — Expotecnológica 2026 (XXI SIE–2026): “Breast lesion classification using deep learning on magnetic resonance imaging”, co-authored with Rubén Darío Fonnegra.

SELECTED PROJECTS

JARVIS — Personal AI Assistant Platform

Modular AI system with a FastAPI backend and a React dashboard.

- LLMs with automatic fallback: a 6-model chain (GPT-OSS 120B/20B and Qwen on Groq, Gemini, local Mistral and Phi-3 with Ollama) behind an in-house gateway with token control, retries and circuit breaker.
- Orchestrator with 4 specialized agents (developer, security, reviewer and drafter), function calling, risk-based permissions, human approval and an audit log of every tool call.
- RAG on Pinecone with multilingual-e5 embeddings and a BGE reranker, validated with a custom evaluation set (recall@3 = 1.0).
- Observability across 9 Docker Compose services (backend, frontend, Redis, Career Ops, Prometheus, Grafana, Loki, Promtail and Jaeger): backend and Career Ops metrics in Prometheus with 6 alerts (service down, 5xx errors, failing tools, action awaiting approval and LLM rate limit), Grafana dashboards, centralized logs with Loki and Promtail, and distributed tracing with Jaeger.
- Career Ops microservice with Redis; Gmail, Google Calendar and Notion integrations; voice with Whisper and Edge TTS; Supabase authentication (JWT).
- 1,000+ automated tests and CI on GitHub Actions; model tracking with MLflow; deployment with Docker and on a university Kubernetes cluster.

FastAPI · React · Claude · Groq · Ollama · Pinecone · Redis · Supabase · MLflow · Prometheus · Grafana · Docker · Kubernetes · Loki · Jaeger · Docker Compose

Lector de PDF — audiobook and English-learning PWA

lector-pdf-pandora.vercel.app

Installable web app (PWA) that turns PDFs, EPUBs, Word files, web pages and photos of books into audiobooks with neural voices running on the device itself, with no accounts or server, and teaches English.

- Piper neural voices running in the browser with ONNX Runtime Web; automatic OCR of scanned pages with Tesseract.js.
- Offline Spanish ⇄ English translation (Firefox Translations in WebAssembly), bilingual reading and spaced-repetition practice (Leitner).

TypeScript · Preact · Vite · PWA · ONNX Runtime Web · Piper · pdf.js · Tesseract.js · WebAssembly · Cloudflare Workers · Playwright · Vitest · GitHub Actions · Umami · Vercel

Breast Cancer Diagnosis Support Tool — Project Lead

Five-person team project for the Intermediate AI Diploma at Universidad de Antioquia (2026), which I led: 10 supervised classifiers on Breast Cancer Wisconsin Diagnostic held to a clinical standard of recall ≥ 0.90 for the malignant class, deployed on AWS.

- Coordinated the team and the repository: two models per member, shared metrics (accuracy, precision, recall, F1, ROC-AUC, false negatives and positives) and consolidated results.
- Author of the EDA that selected 6 features for discriminative power and low multicollinearity, and of the comparative review of the 10 models.

Python · Scikit-learn · Random Forest · SVM · GridSearchCV · Pandas · AWS · EC2 · S3 · VPC · IAM · Docker

Fruit Classification with CNNs

Computer vision with CNNs: evaluation of several architectures, overfitting analysis, MLflow tracking and a reproducible environment.

Predictive Models for Seismic and Fintech Analysis
Time series, fuzzy logic and Monte Carlo simulation applied to seismic and financial analysis, with statistical analysis, feature selection and cross-validation.

TECHNICAL SKILLS

Generative AI & agents: LLMs, Claude, GPT-OSS, Llama 3, Mistral 7B, Phi-3, RAG, Embeddings, Reranking, Pinecone, Qdrant, Prompt engineering, AI agents, Multi-agent systems, Function calling, Local inference with Ollama, Whisper, Claude Code

Machine Learning & Deep Learning: Python, PyTorch, TensorFlow, Keras, Scikit-learn, OpenCV, Computer vision, Medical imaging, CNN, RNN, Transformers, Transfer learning, Grad-CAM

Classic ML models: Linear regression, Logistic regression, KNN, Decision trees, Random Forest, SVM, Naive Bayes, K-Means, Gradient Boosting

Backend & Java: Java 8/12, Spring Boot, Spring Security, JPA, REST APIs, SOAP APIs, Microservices, JSP, Servlets, JDBC, Apache Tomcat, Node.js, Express, FastAPI, Flask

Frontend & web: JavaScript, TypeScript, React, Next.js, Tailwind CSS, HTML5, CSS, Bootstrap, WordPress, Moodle, Google Apps Script

Architecture & design: Hexagonal architecture, Clean Architecture, Layered architecture, SOLID principles, Object-oriented programming, Design patterns, ADRs

Data science: Pandas, NumPy, Matplotlib, Seaborn, Jupyter, Time series, Multivariate regression, Fuzzy logic, Monte Carlo simulation, Statistical analysis, Anomaly detection, Power BI, Risk analysis, Parquet

MLOps, cloud & infrastructure: MLflow, Databricks, PySpark, Apache Spark, Terraform, TensorBoard, Hadoop, Airflow, ETL pipelines, Docker, Kubernetes, AWS (EC2, S3, VPC, RDS), GitHub Actions, Prometheus, Grafana, Vercel

Databases: PostgreSQL, MSSQL, Oracle, MySQL, MongoDB, Redis, Supabase, Flyway

Automation & testing: Playwright, Selenium, Pytest, BeautifulSoup, n8n, Zapier, Make, RPA

Tools & practices: Git, GitHub, GitLab, Postman, Jira, Scrum

EDUCATION

B.Sc. in Software Engineering — Institución Universitaria Pascual Bravo — Feb 2025 — Dec 2026
Coursework completed; graduation expected in December 2026.

Artificial Intelligence Diploma · Intermediate (280 h) — Universidad de Antioquia + Talento Tech — Feb 2026 — Apr 2026

Cloud Architecture Diploma · Advanced (232 h) — ESUMER — Oct 2024 — Dec 2024

Web Programming — Universidad Pontificia Bolivariana — Oct 2022 — Dec 2022

Data Science Diploma (380 h) — ESAP — Aug 2022 — Oct 2022

Software Development Technologist — Institución Universitaria Pascual Bravo — Aug 2017 — Jan 2022

Computer Maintenance Technician — SENA — 2017 — 2018

Process Automation Specialization (36 h) — Nodo + Accenture — Sep 2026 — Nov 2026
In progress: n8n, Zapier, Make, LLM APIs and RAG chatbots.

Digital Transformation with AI and Automation (certified) — Nodo + Accenture Scholarships — 2026 · cohort G3
AI and automation applied to digital transformation: process analysis and identification of automation opportunities.

All of my education has been funded by scholarships earned through good academic performance: Sapiencia, Participatory Budget, Matrícula Cero, Nodo + Accenture and Estud-IA.

CERTIFICATIONS AND AWARDS

- COPNIA professional license — Professional Registration Certificate (Auxiliary).
- 3rd place — Advanced Cloud Architecture Hackathon, ESUMER (Dec 2024): AWS infrastructure with Terraform.

LANGUAGES

Spanish — native · English — B1+ (intermediate)

EARLIER EXPERIENCE · TECHNICAL SUPPORT AND CUSTOMER SERVICE

Multi-Skill Agent — Technical support, sales and retention — OneLink BPO — Aug 2019 — Mar 2021

Technical Support Representative — Emtelco — Jan 2018 — Jan 2019

IT Maintenance Intern — Espumas Medellín — May 2017 — Jan 2018

Customer Service Representative — Teleperformance — Mar 2014 — Nov 2014